

BIOINFORMATICS FOR MS DATA SCIENCE

Elective Course — 32 Lectures / 48 Contact Hours

Recommended course title:

Bioinformatics For Data Science

Level: MS Data Science

Course type: Elective

Duration: 16 Weeks

Lectures: 32

Contact hours: 48

Lecture duration: 1.5 hours

Good to Know for This Course: Python, statistics/probability, basic machine learning, basic linear algebra, introductory biology

1. Course Description

Bioinformatics is an interdisciplinary field combining biology, computer science, statistics, data science, artificial intelligence, and computational modeling to analyze biological and biomedical data.

This course introduces MS Data Science students to modern computational biology with an emphasis on real biological datasets and data-science workflows.

Students will learn how to:

- obtain biological datasets from public repositories;
- process DNA, RNA, and protein sequences;
- perform sequence alignment and similarity searches;
- analyze next-generation sequencing data;
- perform genome assembly and annotation;
- identify genetic variants;
- analyze RNA-seq and single-cell RNA-seq;
- perform differential expression and enrichment analysis;
- analyze biological networks;
- apply machine learning to biological problems;
- use AI for protein structure and function prediction;
- work with long-read sequencing and pangenomes;
- analyze multi-omics data;
- build reproducible bioinformatics pipelines;
- use cloud/HPC resources;
- interpret biological results using appropriate statistics and visualization.

The course deliberately connects bioinformatics concepts with Data Science concepts such as preprocessing, feature engineering, dimensionality reduction, clustering, classification, deep learning, graph learning, reproducibility, and model evaluation.

2. Course Learning Outcomes

By the end of the course, students should be able to:

CLO 1 — Biological Data Understanding

Explain major biological data types including:

- DNA
- RNA
- proteins
- genomes
- transcriptomes
- epigenomes
- microbiomes
- single-cell data
- spatial omics
- multi-omics

CLO 2 — Biological Databases

Use major resources such as:

- NCBI
- SRA
- GEO
- GenBank
- Ensembl
- UCSC Genome Browser
- UniProt
- PDB
- AlphaFold DB
- ClinVar
- dbSNP
- gnomAD

NCBI's SRA, for example, provides access to large-scale publicly available sequencing datasets and associated analysis tools.

CLO 3 — Sequence Analysis

Apply:

- sequence alignment;
- BLAST;
- pairwise alignment;
- multiple sequence alignment;
- phylogenetic analysis;

- motif discovery.

BLAST remains an important sequence-similarity tool, with nucleotide/protein searches, specialized searches, APIs and databases.

CLO 4 — Genomics

Understand and implement basic:

- NGS preprocessing;
- read alignment;
- genome assembly;
- variant calling;
- variant annotation;
- genomic visualization.

CLO 5 — Transcriptomics

Perform computational analysis of:

- RNA-seq;
- differential expression;
- pathway enrichment;
- single-cell RNA-seq.

Current Bioconductor material includes modern workflows for single-cell RNA-seq analysis, reflecting the importance of cell-level transcriptomic analysis in contemporary bioinformatics.

CLO 6 — Machine Learning

Apply:

- supervised learning;
- unsupervised learning;
- dimensionality reduction;
- deep learning;
- graph-based methods;
- representation learning

to biological datasets.

CLO 7 — AI for Biology

Explain and use modern AI approaches including:

- protein language models;
- genomic language models;
- protein structure prediction;
- AlphaFold;
- AlphaFold 3;

- AI-based variant-effect prediction;
- biological foundation models.

AlphaFold 3 introduced a diffusion-based architecture capable of modeling complexes involving proteins, nucleic acids, small molecules, ions and modified residues.

CLO 8 — Reproducible Bioinformatics

Develop reproducible pipelines using:

- Git/GitHub;
- Conda/Mamba;
- Docker;
- Nextflow;
- nf-core;
- notebooks;
- workflow metadata.

CLO 9 — Data Interpretation

Critically evaluate biological results using:

- statistical testing;
- effect sizes;
- confidence;
- multiple-testing correction;
- biological validation;
- appropriate visualization.

CLO 10 — Research Project

Complete an end-to-end bioinformatics/data-science project using a real biological dataset.

3. Course Philosophy

The course should not become a memorization-based biology course.

The central question should be:

"How can a data scientist transform biological data into reliable biological knowledge?"

4. Modern Technology Stack

The course should expose students to the following ecosystem.

Programming

- Python

- R
- Bash/Linux
- SQL where appropriate
- Jupyter Notebook

Python ecosystem

- NumPy
- pandas
- SciPy
- Biopython
- scikit-learn
- matplotlib
- seaborn
- PyTorch
- Scanpy
- AnnData

R/Bioconductor

- Bioconductor
- DESeq2
- edgeR
- limma
- Seurat
- SingleCellExperiment
- scater
- scran

Sequence analysis

- BLAST
- MAFFT
- MUSCLE
- HMMER
- CD-HIT

Genomics

- FastQC
- MultiQC
- Fastp
- BWA
- Bowtie2
- minimap2
- samtools
- bcftools
- GATK

- DeepVariant

GATK's current Best Practices cover germline and somatic short variants, CNVs, structural variation and preprocessing workflows.

Genome visualization

- IGV
- UCSC Genome Browser
- Ensembl

UCSC currently provides genome visualization, BLAT, Table Browser, LiftOver, REST APIs and variant annotation tools, and supports current assemblies including human T2T-CHM13.

Transcriptomics

- STAR
- Salmon
- kallisto
- DESeq2
- edgeR
- limma

Single-cell

- Scanpy
- Seurat
- scVI
- Cell Ranger concepts
- AnnData
- CellxGene
- Bioconductor single-cell ecosystem

Long-read sequencing

- minimap2
- samtools
- Flye
- Canu
- hifiasm
- Dorado
- Clair3
- Longshot

Pangenomics

- Minigraph
- Minigraph-Cactus
- VG

- Giraffe
- pangenome graphs

Pangenome analysis is increasingly important because a single reference genome can introduce representation bias; graph-based pangenomes allow genomic variation from multiple individuals to be represented together.

Protein bioinformatics

- BLASTp
- UniProt
- PDB
- AlphaFold DB
- AlphaFold 2
- AlphaFold 3
- ColabFold
- ESM/Protein Language Models

Workflow engineering

- Nextflow
- nf-core
- Snakemake
- Docker
- Apptainer/Singularity
- Conda/Mamba

Computing

- Google Colab
- HPC clusters
- cloud computing
- GPU computing
- object storage
- workflow orchestration